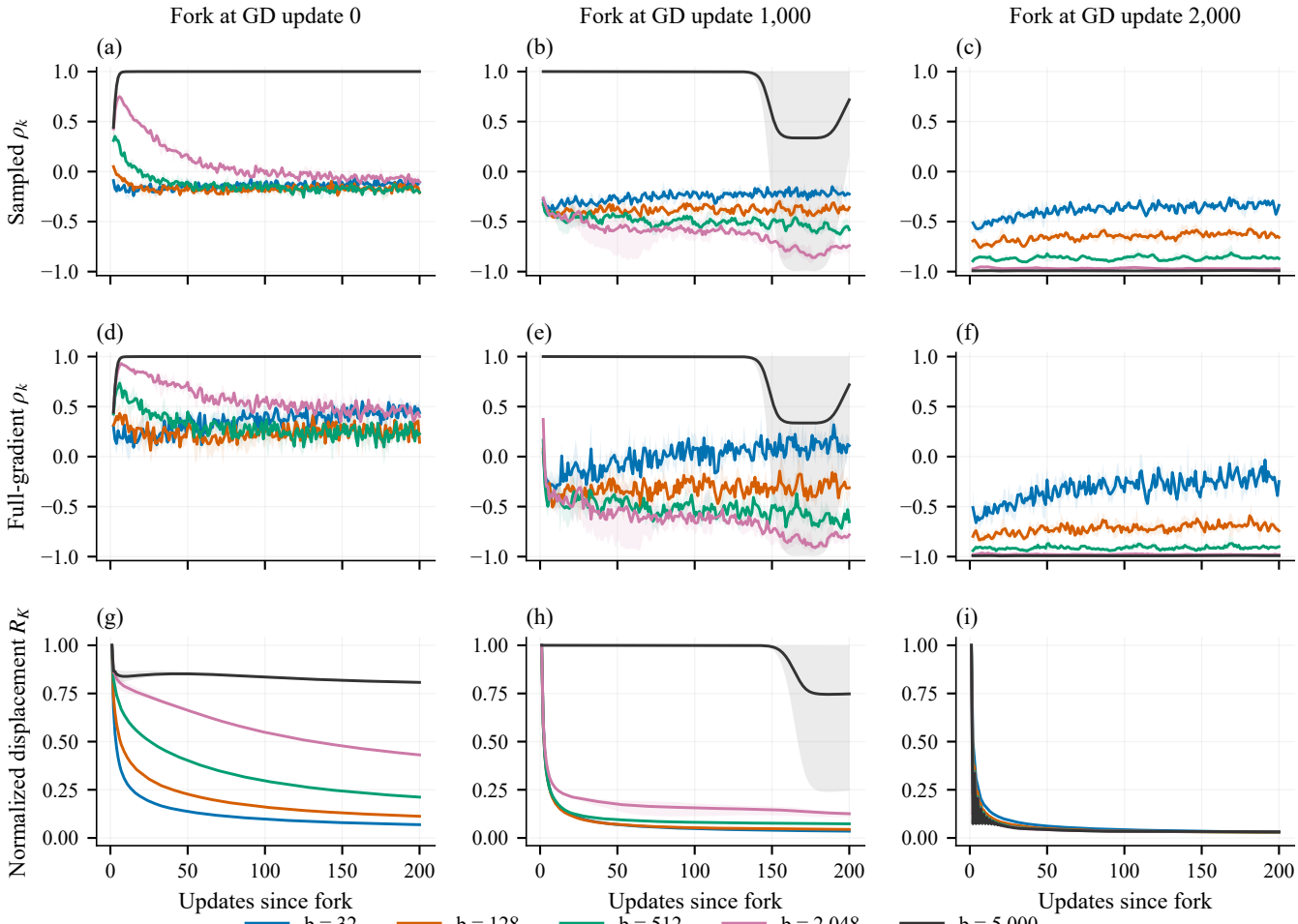

Fork at GD update 0
Fork at GD update 1,000
Fork at GD update 2,000
(a)
(b)
(c)
(d)
(e)
(f)
(g)
(h)
(i)
Sampled ρ_k
Full-gradient ρ_k
Normalized displacement R_K
1.0
0.5
0.0
−0.5
−1.0
1.00
0.75
0.50
0.25
0.00
0
50
100
150
200
Updates since fork
b = 32
b = 128
b = 512
b = 2,048
b = 5,000